

**PARKTOWN BOYS' HIGH SCHOOL***Department of Physical Science*

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**Prelim Exam**

13 September 2024

|                |                  |              |         |
|----------------|------------------|--------------|---------|
| <b>Subject</b> | Physical Science | <b>Marks</b> | 150     |
| <b>Grade</b>   | 12               | <b>Time</b>  | 3 hours |
| <b>Topic</b>   | Chemistry        |              |         |

**GENERAL INSTRUCTIONS**

- 1. Answer ALL the questions on your question paper.*
- 2. Read the particular instructions for each section carefully.*
- 3. Answer each question on a NEW PAGE.*
- 4. This test consists of 18 pages which includes this cover page and a formula sheets.*

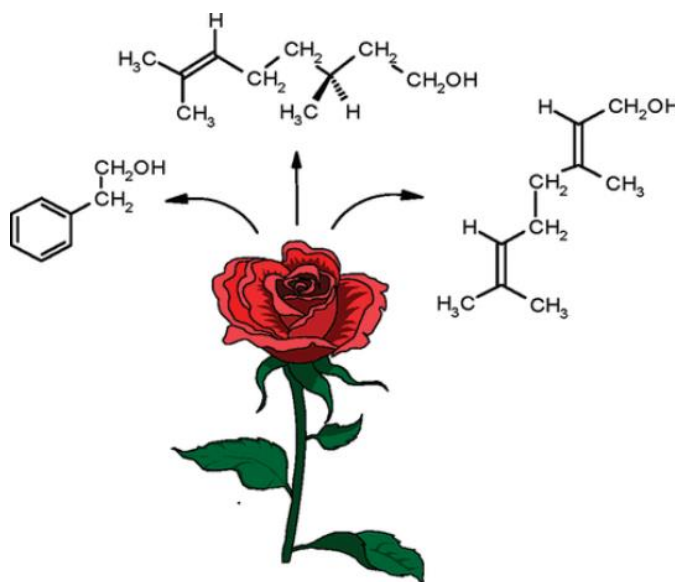
|                           |               |
|---------------------------|---------------|
| <b>Examiner</b>           | Ms L Whiting  |
| <b>Moderator</b>          | Mrs I Pelsner |
| <b>External moderator</b> | Mrs Patel     |

**Question 1****Multiple choice****[20]**

1.1 If a scientific experiment yields precise results, it means that ...

- A the measuring equipment was correctly calibrated.
- B all variables other than the independent variable were fixed.
- C similar results will be obtained when the experiment is repeated.
- D the average of the repeat results is close to the accepted value.

1.2 The scent of a red rose consists of a few different molecules represented by the structural formulae given below.



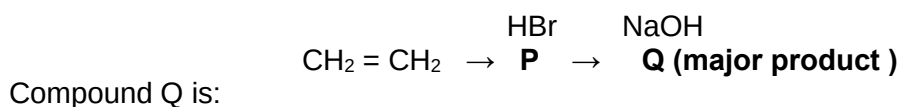
The structural formulae above are examples of the following homologous series:

- A haloalkane, ester
- B alcohol, alkene
- C carboxylic acids
- D alcohol, alkane

1.3 The melting point of a carboxylic acid is high as a result of ...

- A strong polar covalent bonds between atoms.
- B strong electrostatic forces between ions.
- C strong intermolecular forces between molecules.
- D strong ion-dipole forces between the ions.

1.4 In the flow diagram below **P** and **Q** represent two organic compounds



- A  $\text{CH}_2\text{CH}_2$
- B  $\text{CH}_3\text{CH}_3$
- C  $\text{CH}_3\text{CH}_2\text{Br}$
- D  $\text{CH}_3\text{CH}_2\text{OH}$

1.5 In the reaction  $\text{Zn (s)} + \text{HCl (aq)} \rightarrow \text{ZnCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$ , the rate of reaction can be reduced by:

- A Adding more Zn.
- B Adding water to the mixture.
- C Adding a catalyst .
- D stirring the mixture

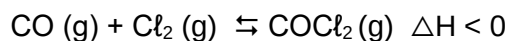
1.6 Sacrificial anodes are used to protect other metals that oxidise easily. Which one of the following metals would protect the anchor of a ship from oxidation? The anchor is made of iron that will oxidise into  $\text{FeO}$ .

- A. Fe(III)
- B. Ni
- C. Sn
- D. Mg

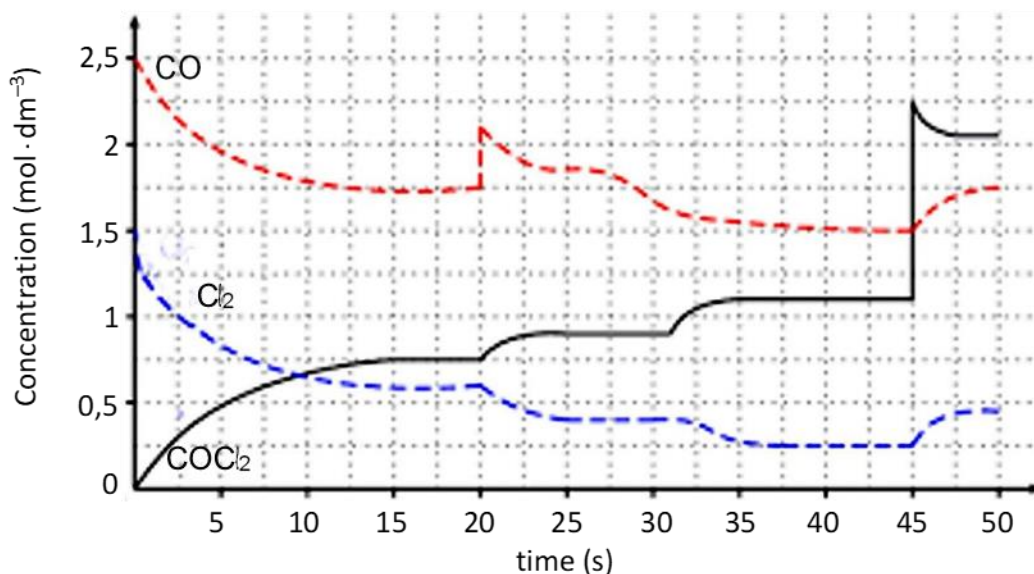
1.7 Which ONE of the following is a NON-SPONTANEOUS redox reaction?.

- A.  $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
- B.  $\text{Cu(s)} + \text{FeCl}_2\text{(aq)} \rightarrow \text{CuCl}_2\text{(aq)} + \text{Fe(s)}$
- C.  $2\text{AgNO}_3\text{(aq)} + \text{Cu(s)} \rightarrow \text{Cu(NO}_3)_2\text{(aq)} + 2\text{Ag(s)}$
- D.  $2\text{Al(s)} + 3\text{Ni(NO}_3)_2\text{(aq)} \rightarrow 2\text{Al(NO}_3)_3\text{(aq)} + 3\text{Ni(s)}$

1.8 The balanced equation below represents a reaction that occurs in a sealed container:



Various changes were made to the reaction conditions at various times as shown by the graphs below:



Which of the following statements regarding the changes made to the equilibrium mixture are true?

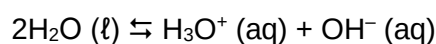
- (i) at 20 s the concentration of carbon monoxide gas was increased
- (ii) at 32,5 s the pressure of the system was increased
- (iii) at 32,5 s the temperature was decreased suddenly

- A i, ii and iii
- B i only
- C i and iii only
- D ii and iii only

1.9 Consider the salt  $\text{MgSO}_4$  and write down the letter that will explain the pH of the salt solution correctly.

- A. **Basic:**  $\text{Mg}(\text{OH})_2$  is a strong base. Amount of  $\text{OH}^-$  more than  $\text{H}_3\text{O}^+$  as strong base ionises completely but weak acid,  $\text{H}_2\text{SO}_4$ , ionises partially.
- B. **Neutral:**  $\text{Mg}(\text{OH})_2$  is a strong base. Amount of  $\text{OH}^-$  is equal to  $\text{H}_3\text{O}^+$  as strong base ionises completely and strong acid,  $\text{H}_2\text{SO}_4$ , fully ionises.
- C. **Acidic:**  $\text{Mg}(\text{OH})_2$  is a weak base. Amount of  $\text{OH}^-$  less than  $\text{H}_3\text{O}^+$  as weak base partially ionised but strong acid,  $\text{H}_2\text{SO}_4$ , fully ionises.
- D. **Acidic:**  $\text{Mg}(\text{OH})_2$  is a weak base. Amount of  $\text{H}_3\text{O}^+$  less than  $\text{OH}^-$  as weak base partially ionised but strong acid,  $\text{H}_2\text{SO}_4$ , fully ionises.

1.10 Water undergoes auto-protolysis and sets up the following equilibrium:



Which of the following statements is **incorrect**:

- A. Both the forward and reverse reactions proceed at the same rate. B Most of the water molecules do not ionise
- C. The product of the concentrations of the hydronium and hydroxide ions always remains constant at a given temperature.
- D. The equilibrium constant will be  $K_w = 1 \times 10^7$  as water is not acidic nor basic.

**Question 2****Organic Molecules****[19]**

Use the following table to answer the following questions.

|          |                                                                                                                                                                                                                                                                                 |          |                                                                                                                                                                                                                                        |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | $  \begin{array}{c}  \text{CH}_3 \quad \quad \text{H} \\  \diagdown \quad \diagup \\  \text{C} = \text{C} \\  \diagup \quad \diagdown \\  \text{CH}_3 \quad \quad \text{CH}_3  \end{array}  $                                                                                   | <b>B</b> | $  \begin{array}{c}  \text{H} \quad \quad \quad \text{H} \\    \quad \quad \quad   \\  \text{H}-\text{C}-\text{C}\equiv\text{C}-\text{C}-\text{H} \\    \quad \quad \quad   \\  \text{H} \quad \quad \quad \text{CH}_3  \end{array}  $ |
| <b>C</b> | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$                                                                                                                                                                                                                         | <b>D</b> | 2,2 dimethylpropane                                                                                                                                                                                                                    |
| <b>E</b> | $  \begin{array}{ccccccc}  & \text{H} & & \text{O} & & \text{H} & \text{H} \\  &   & &    & &   &   \\  \text{H}- & \text{C} & -\text{O}- & \text{C} & - & \text{C} & - & \text{C} & -\text{H} \\  &   & & & &   &   \\  & \text{H} & & & & \text{H} & \text{H}  \end{array}  $ | <b>F</b> | $  \begin{array}{c}  \text{O} \\     \\  \text{CH}_3\text{CH}_2\text{CCHCH}_3  \end{array}  $                                                                                                                                          |

[Source: <<https://megatude.co.za/wp-content/uploads/2018/08/Physical-Science-Grade-12-Definitions-Paper-2-Free-State.pdf>>]

2.1 Define intermolecular force. (2)

2.2 Identify the LETTER that represents the following: (3)

2.2.1 Alkyne

2.2.2 Alcohol

2.3.2 Name the positional ISOMER of C ompound E

2.3 Compound C reacts with reacts with a carboxylic acid to form compound S in the presence of a strong acid.

2.3.1 Name the type of reaction that takes place. (1)

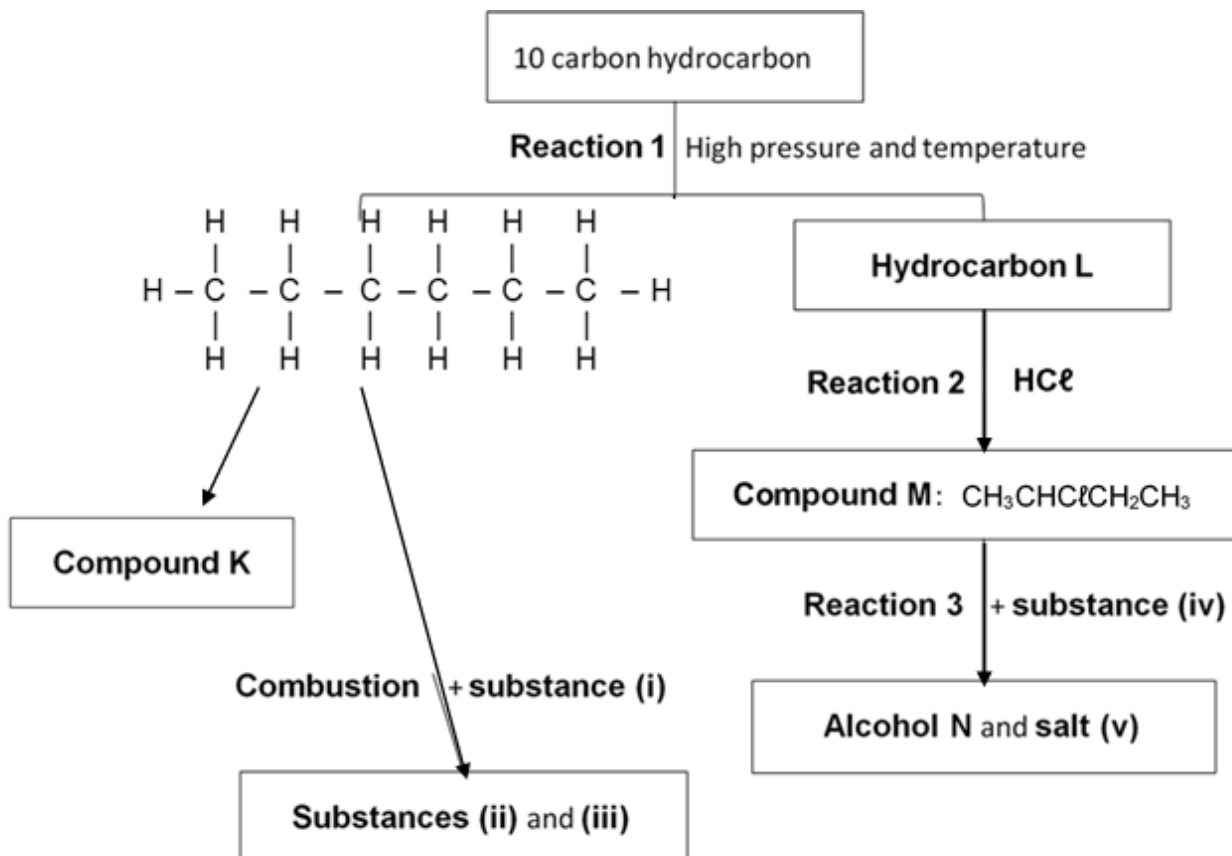
2.3.2 NAME or FORMULA of the inorganic acid catalyst (1)

2.3.3 Property of alcohols that make it necessary to use a water bath to heat the reaction mixture instead of direct heat (1)

- 2.4. Compound S has an EMPIRICAL FORMULA of  $C_3H_6O$  and a molecular mass of  $116 \text{ g}\cdot\text{mol}^{-1}$ . Write down the MOLECULAR FORMULA of the carboxylic acid. (3)
- 2.5 Give the IUPAC name of Compound B. (2)
- 2.6  $C_4H_{10}O$  is a tertiary alcohol. Write down the STRUCTURAL FORMULA and IUPAC name of compound C. (4)
- 2.7 Write down the MOLECULAR FORMULAE of the TWO products formed during the complete combustion of compound **D**. (2)

**Question 3****[18]**

Consider the following flow diagram for several organic reactions:



- 3.1 For reaction 1, identify both the GENERAL type of reaction and SPECIFIC type of reaction that have occurred. (2)
- 3.2 Write the balanced combustion reaction for hexane hence identifying substances (i)–(iii). Use molecular formulae. (3)
- 3.3 Compound K is a structural isomer of hexane.
- 3.4 Write down the condensed structural formula of a possible isomer of hexane. (2)
- 3.5 Give the IUPAC name for hydrocarbon L. (1)
- 3.6 Refer to reaction 2.
  - 3.6.1 Identify the SPECIFIC type of reaction that took place. (1)
  - 3.6.2 Write down one reaction condition for this reaction to take place. (1)



3.7 Refer to reaction 3 which produces alcohol N and a salt.

3.7.1 Identify the GENERAL type of reaction that took place. (1)

3.7.2 Draw the structural formula for alcohol N. (3)

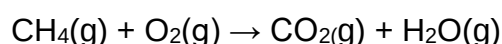
3.7.3 Identify substance (iv) and salt (v). (2)

3.7.4 State the reaction conditions required for this reaction to occur. (2)

**Question 4****[15]**

The largest and most powerful rocket ever built has blasted off from Texas but blew up within minutes in a test flight that its makers, Space X, hoped will be the first step in the journey to Mars. After a cancelled launch because of a pressurisation issue, the 120-metre Starship rocket system took off, but as it gathered speed, it started to spin at an altitude of 39 kilometres before exploding about four minutes after leaving the ground (The Guardian, 20 April 2023).

The explosion reaction that takes place is represented by the chemical equation shown below.

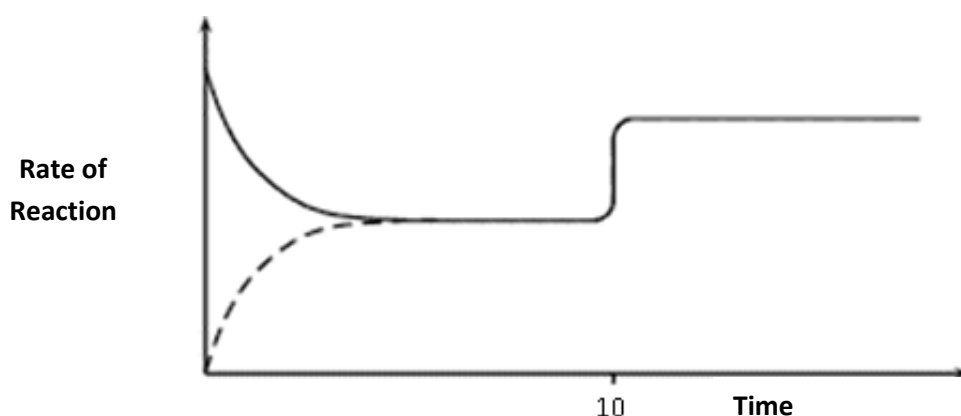


4.1 Define an exothermic reaction. (2)

4.2 In your answer booklet, sketch a labelled potential energy graph of the reaction above. (5)

4.3 Elon Musk stated that the explosion was intentional. The engineers initiated the reaction between the methane and oxygen gas. Use the collision theory and explain how increasing the amount of oxygen gas in the chamber will affect the reaction rate. (4)

4.4 The following reaction rate–time graph was obtained from the reaction.



[Source: <<https://www.siyavula.com/read/za/physical-sciences/grade-12/chemical-equilibrium/08-chemical-equilibrium-03>>]

4.4.1 Identify the change in the reaction at 10 min to obtain this graph if atmospheric

pressure remained the same.

(1)

4.4.2 Explain your answer given in Question 4.4.1. and write whether the amount of product formed will INCREASE, DECREASE or REMAIN THE SAME. (3)

## Question 5 Chemical Equilibrium

[19]

5.1 The decomposition reaction of calcium carbonate is reversible in a closed system and reaches an equilibrium at 866°C, in the presence of a catalyst.



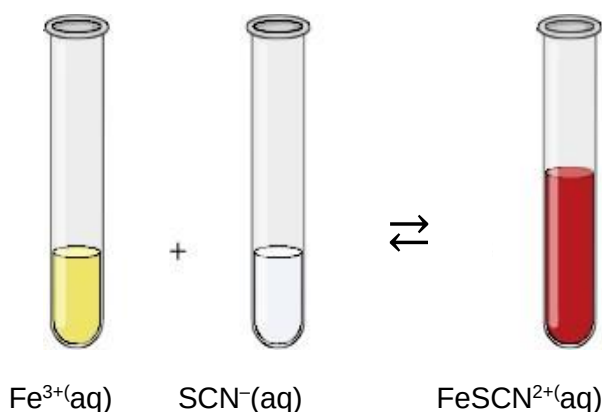
5.1.1 Define a closed system. (2)

5.1.2 State Le Chatelier's Principle. (2)

5.1.3 Write the K<sub>c</sub> expression for this reaction. (2)

5.1.4 Initially 35 g of CaCO<sub>3</sub> was put in the closed container with a volume of 500 cm<sup>3</sup> and the temperature increased to 866°C. At equilibrium 23 g CaCO<sub>3</sub> was present in the system. Calculate the equilibrium constant. (5)

5.2 When pale yellow iron(III) ions are added to a colourless solution of thiocyanate ions (SCN<sup>-</sup>), a blood red colour develops rapidly as the thiocyanateiron(III) ion is formed. The test tube also feels warm. The colour is caused by a complex ion with the formula (FeSCN<sup>2+</sup>).

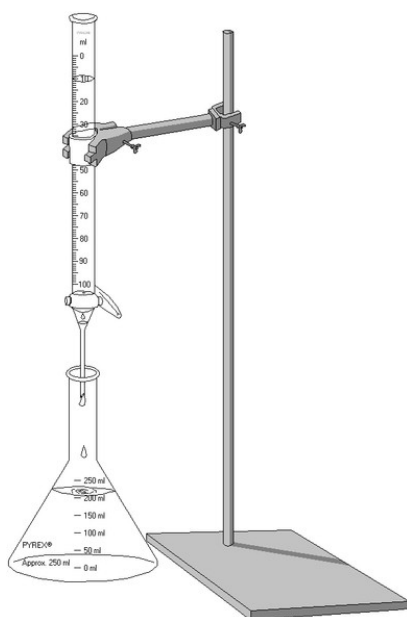


[Source: <<https://www.nagwa.com/en/explainers/194142843531/>>]

- 5.2.1 According to Le Chatelier, explain the effect that an increase in temperature would have on the reaction. (4)

**Question 6**                      **Acids and Bases**                      **may 2016**                      **[22]**

A student titrated a solution of sodium hydroxide with a standard solution of hydrochloric acid. The apparatus used during the experiment is shown below.



[Source: <[https://commons.wikimedia.org/wiki/File:Titration\\_Apparatus.png](https://commons.wikimedia.org/wiki/File:Titration_Apparatus.png)>]

The following table of indicators might be useful in the questions that follow.

| Indicator        | Acidic Colour | pH Range of Colour Change | Basic Colour |
|------------------|---------------|---------------------------|--------------|
| Phenolphthalein  | Colourless    | 8,3–10,0                  | Pink         |
| Bromothymol blue | Yellow        | 6,0–7,6                   | Blue         |
| Phenol red       | Yellow        | 6,8–8,0                   | Red          |
| Methyl orange    | Red           | 3,2–4,4                   | Yellow       |

[Source: <<https://scienceready.com.au/pages/ph-indicators>>]

- 6.1 Explain why hydrochloric is a strong acid.

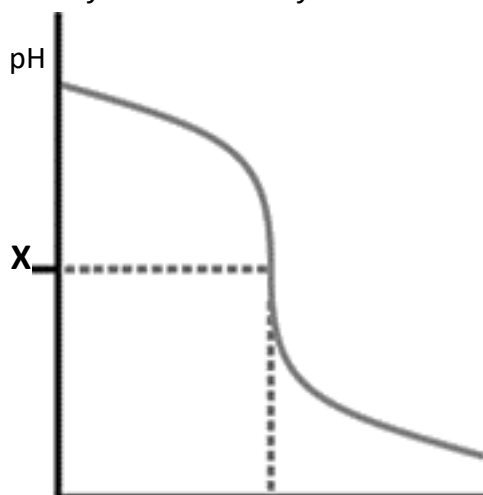
- 6.2 During the titration, a standard solution of hydrochloric acid reacts with a sodium hydroxide solution with a concentration of  $0,2 \text{ mol}\cdot\text{dm}^{-3}$ . The chemical equation representing this titration is given below:



Assuming that the sodium hydroxide fully dissociates in the solution, calculate the concentration of hydronium ions in the standard solution of hydrochloric acid. (4)

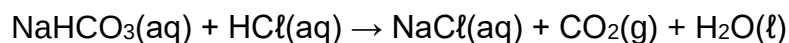
- 6.3 The curve below shows the change in pH observed in the Erlenmeyer flask during the above titration.

Titration curve for sodium hydroxide and hydrochloric acid.



- 6.3.1 Identify point X on the graph above. Give a possible value for point X for this specific titration. (2)
- 6.3.2 Use Le Chatelier's principle and refer to the above equation to explain what colour change will be observed during the titration of sodium hydroxide and hydrochloric acid. Your explanation must include the initial colour, the neutralised colour and the over-titrated colour. (4)

- 6.4 During another experiment  $0,50 \text{ dm}^3$  of a  $0,10 \text{ mol} \cdot \text{dm}^{-3}$   $\text{HCl}$  solution is added to  $0,80 \text{ dm}^3$  of a  $\text{NaHCO}_3$  solution of concentration  $0,25 \text{ mol} \cdot \text{dm}^{-3}$ . The balanced equation for the reaction is:

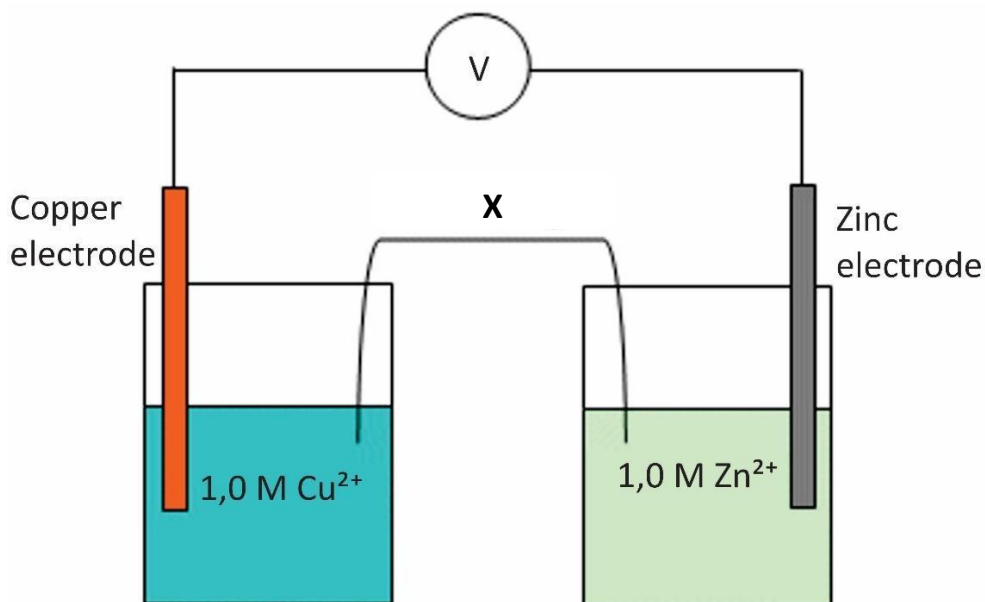


Calculate the:

- 6.4.1 Concentration of the hydroxide ions in the solution on completion of the reaction (8)
- 6.4.2 pH of the solution on completion of the reaction (4)
- 6.4.3 Give the autoionisation reaction equation of water and indicate the acid-base pairs. (4)

**Question 7 Galvanic Cells****[20]**

An electrochemical cell is constructed as shown. Study the diagram drawn below and answer the questions that follow.



[Source: <<http://www.4college.co.uk/a/ss/cell.php>>]

7.1 Consider the aqueous copper nitrate solution as seen above.

7.1.1 Give the formula of the solute that was used to make the copper(II)nitrate solution. (2)

7.1.2 Name the solvent that was used to make the electrolyte. (1)

7.2 The zinc rod is now replaced by an iron rod, and the zinc solution is replaced with an iron(II)nitrate solution.

7.2.1 Name the reducing agent in this reaction. (1)

7.2.2 Give the equations for the oxidation and reduction half reactions. (2)

7.2.3 Write the cell notation for this cell. (4)

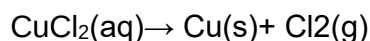
7.2.4 State which electrode, Cu or Fe, would decrease in mass as the reaction proceeds. Explain your answer. (3)

7.2.5 Would a potassium chloride solution be a suitable chemical to use in component X? Explain your answer by mentioning the movement of the different ions. (4)

7.2.6 Calculate the emf generated by this cell under standard conditions. (3)

## **Question 8                      Electrolytic Cells                      [18]**

8.1 A student wishes to observe the electrolysis of a solution of copper(II) chloride. He inserts two graphite electrodes into the solution and connects these electrodes to a DC power supply. The following balanced equation shows the predominant reaction that occurs:



8.1.1 Explain why the student uses copper(II) chloride solution and not copper(II) chloride solid for this process. (3)

8.1.2 At which electrode, anode or cathode, will chlorine gas be produced? (2)

8.1.3 The student adds copper(II) chloride to the cell every minute to prevent the formation of oxygen gas at the electrode named in Question 8.1.2. Why does the student do this? Fully explain your answer. (3)

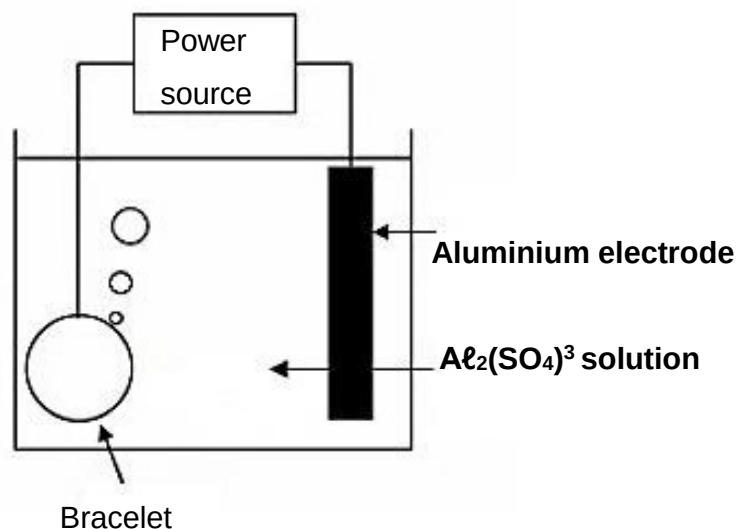
After a period of 10 minutes, 0,11 mol of copper metal is formed.

8.1.4 Determine the charge supplied to this cell in 10 minutes. (4)

8.1.5 Calculate the number of chloride ions used while the cell runs for 10 minutes. (3)



8.2 In an attempt to electroplate a bracelet with aluminium metal, a technician uses an aqueous solution of aluminium sulfate in a cell as shown below.



When the technician connects the power source, she notices that there is no shiny plating of aluminium metal appearing on the bracelet. Instead, there is a stream of colourless bubbles forming identified as hydrogen gas due to the popping sound it makes when a lighter is lit above the bracelet.

8.2.1 Over time, what will be observed at the aluminium electrode? (1)

8.2.2 Name the gas produced at the bracelet. Explain your answer. (2)

Annexure 1.

Question 5.2

